Probability & Statistics Exam Guide

Standard PMFs/PDFs and many identities are on the supplied sheet. This guide emphasizes meaning, recognition, setup and reusable exam patterns.

0. Start every subpart

Define the RV/event in words; write its **support**; identify the model or theorem and its assumptions; set parameters with units; translate the requested event exactly; compute; then check range, normalization and limiting sense.

$$\mu = E[X], \qquad \sigma^2 = \operatorname{Var}(X), \qquad \sigma = \sqrt{\operatorname{Var}(X)}.$$

Useful translations: "at least one" $= 1 - P(0)$; $T_k \le t \iff N(t) \ge k$; $\max_i X_i \le t$ means every $X_i \le t$; $\min_i X_i > t$ means every $X_i > t$.

1. Conditioning, total probability and Bayes

Conditioning changes the sample space to the information already known. Thus $P(A \mid B)$ asks what fraction of the remaining B-world also satisfies A.

Lookup: הסתברות מותנה

Which conditioning tool? "Given that" → condition directly. A hidden case, random size or random composition → solve within each case and average. Reversing evidence and cause → Bayes. A simple conditional distribution but awkward marginal → total expectation.

$$P(A) = \sum_h P(A \mid H = h)P(H = h),$$
$$E[X] = E(E[X \mid H]).$$

Lookup: נוסחת ההסתברות השלמה *Lookup:* משפט התוחלת השלמה

Bayes weights each possible cause by "prior × likelihood of the evidence" and normalizes:

$$P(H_j \mid A) = \frac{P(A \mid H_j)P(H_j)}{\sum_i P(A \mid H_i)P(H_i)}.$$

If $L = W^c$, then within any information event A with $P(A) > 0$,

$$P(L \mid A) = 1 - P(W \mid A).$$

Trap: Do not complement only the numerator; the complement is taken inside the conditioned sample space.

2. Counting and choosing a discrete model

Counting first *Lookup:* קומבינטוריקה Use favorable/total only when elementary outcomes are equally likely. Ask: does order matter? replacement? must objects be together or separated? "At least one" is often a complement; "every type appears" often inclusion–exclusion.

Distribution recognition *Lookup:* התפלגויות בדידות

- One yes/no trial → Bernoulli.
- Successes in a *fixed* number of independent trials with constant p → $\mathrm{Bin}(n, p)$.
- Sample without replacement from a finite population → $\mathrm{HG}(N, D, n)$.
- Trials through first / rth success → $\mathrm{Geo}(p)$ / $\mathrm{NB}(r, p)$; the successful trial is counted.
- Events in fixed exposure at constant rate → $\mathrm{Poi}(\lambda t)$.
- Every listed value genuinely equally likely → discrete uniform.

Trap: Without replacement is generally not binomial. Binomial fixes trials; negative binomial fixes the required number of successes.

Series that actually occur For $|r| < 1$,

$$\sum_{k=0}^{n} r^k = \frac{1 - r^{n+1}}{1 - r}, \qquad \sum_{k=0}^{\infty} r^k = \frac{1}{1 - r},$$

$$\sum_{k=m}^{\infty} r^k = \frac{r^m}{1 - r}, \qquad \sum_{k=1}^{\infty} k r^{k-1} = \frac{1}{(1 - r)^2}.$$

Use them for random experiment size, geometric tails, truncated waiting times and expectations containing kr^{k-1}.

Random N (2025/2026 pattern). If $N \sim \mathrm{Geo}(q)$ and winning given $N = n$ requires n independent wins of probability s, then

$$P(W) = \sum_{n \ge 1} q(1 - q)^{n-1} s^n = \frac{qs}{1 - (1 - q)s}.$$

Condition first; do not replace N by $E[N]$.

Once $p_0 = P(\text{win one complete game})$ is known, playing independent full games until the rth win gives $G \sim \mathrm{NB}(r, p_0)$ and $E[G] = r/p_0$.

Tail-sum identity For any nonnegative integer-valued X,

$$E[X] = \sum_{k=1}^{\infty} P(X \ge k).$$

For $Y = \min(X, m)$, truncate the sum at m.

Capped geometric waiting time If $W \sim \mathrm{Geo}(p)$ but the experiment stops after m trials, write $Y = \min(W, m)$. Then for $k \le m$, $P(Y \ge k) = P(W \ge k) = (1 - p)^{k-1}$, so

$$E[Y] = \sum_{k=1}^{m} P(Y \ge k) = \frac{1 - (1 - p)^m}{p}.$$

This avoids listing the full PMF; the mass at m absorbs $P(W \ge m)$.

3. Discrete PMF/CDF and hidden composition

A PMF assigns mass; the CDF accumulates it. *Lookup:* פונקציית ההתפלגות המצטברת For discrete X, $P(a < X \le b) = F(b) - F(a)$ and $P(X > k) = 1 - F(k)$. Always state the support.

Hidden composition → sample (2025). If Y is the random number of type-A items among N, and n items are sampled without replacement, then

$$X \mid Y = y \sim \mathrm{HG}(N, y, n), \qquad E[X \mid Y] = \frac{nY}{N}.$$

Hence $E[X] = nE[Y]/N$ without deriving the unconditional PMF.

A discrete joint table is a map of feasible pairs: impossible cells are zero; row/column sums give marginals; a conditional row is joint divided by the conditioning marginal and must sum to 1.

iid symmetry If independent X, Y have the same distribution,

$$P(Y > X) = P(X > Y) = \tfrac{1}{2}\{1 - P(X = Y)\},$$
$$P(X = Y) = \sum_k P(X = k)^2.$$

4. Indicators, covariance and bounds

Indicators turn a difficult count into a sum of simple 0/1 variables. If $Y = \sum I_i$ with $I_i = \mathbf{1}_{A_i}$, then

$$E[Y] = \sum_i P(A_i)$$

without any independence assumption. *Lookup:* אינדיקטורים For variance, dependence matters:

$$\operatorname{Var}(Y) = \sum_i p_i(1 - p_i) + 2\sum_{i<j} \operatorname{Cov}(I_i, I_j),$$

$$\operatorname{Cov}(I_i, I_j) = P(A_i \cap A_j) - p_i p_j.$$

Classify pairs by overlap, compute one joint probability for each class, and count how many such pairs exist.

Adjacent-comparison pattern (2026). $I_i = \mathbf{1}_{\{X_i < X_{i+1}\}}$ for iid continuous X_i. Then $P(I_i = 1) = 1/2$. Adjacent indicators overlap: $P(X_i < X_{i+1} < X_{i+2}) = 1/6$, so $\operatorname{Cov}(I_i, I_{i+1}) = 1/6 - 1/4 = -1/12$; nonadjacent pairs are independent.

Bounds Markov uses only nonnegativity and a mean: $P(X \ge a) \le E[X]/a$. *Lookup:* אי שוויון מרקוב Chebyshev uses mean and variance: $P(|X - \mu| \ge a) \le \sigma^2/a^2$. *Lookup:* אי שוויון צ'בישב A requested lower bound is often obtained by complementing the upper-tail bound.

Exact, bound or approximation? Use an exact law/integral when it is short. Use Markov or Chebyshev when only moments are available or dependence blocks the exact distribution. Use a normal approximation only for a genuinely large count/sum after checking its assumptions. A bound is not an estimated probability.

Random sum/reward If $S = \sum_{i=1}^{N} W_i$, where the iid rewards are independent of N, conditioning gives

$$E[S] = E[N]E[W],$$
$$\operatorname{Var}(S) = E[N]\operatorname{Var}(W) + \operatorname{Var}(N)E[W]^2.$$

The second term is the variation caused by the random number of rewards.

5. Poisson process: counts, times and structure

Let $N(t)$ count arrivals by time t at rate λ per unit time.

Count or waiting time? Number in interval length t: $N(t) \sim \mathrm{Poi}(\lambda t)$. Time to next arrival: $T \sim \mathrm{Exp}(\lambda)$. Convert minutes/hours before multiplying the rate.

Stationary increments mean only interval length matters; independent increments mean disjoint intervals contribute independent counts. This is why a timeline can be split into independent pieces. Independent Poisson processes superpose: rates add because expected event counts add over every interval. Thinning keeps each event with probability p, giving an independent Poisson process of rate $p\lambda$; different categories are independent. Given $N(T) = n$, the labeled arrival locations are iid $U(0, T)$; the ordered arrival times are their order statistics. Therefore a subinterval occupying fraction a of the full interval contains $\mathrm{Bin}(n, a)$ arrivals. Interarrival times are iid exponential and memoryless. The kth arrival time is a sum of k exponentials, but the event translation

$$T_k \le t \iff N(t) \ge k$$

is often faster than deriving its density. The minimum of independent exponentials with rates λ_i is $\mathrm{Exp}(\sum_i \lambda_i)$; the next event is type j with probability $\lambda_j / \sum_i \lambda_i$.

Thinned revenue (2025). Arrivals have rate λ; type i has probability p_i and reward c_i. Over time t, independent $N_i \sim \mathrm{Poi}(\lambda t p_i)$ and $R = \sum c_i N_i$, so

$$E[R] = \lambda t \sum c_i p_i, \qquad \operatorname{Var}(R) = \lambda t \sum c_i^2 p_i.$$

Square rewards, not probabilities, in the variance.

Poisson count during a random time If $N \mid T = t \sim \mathrm{Poi}(\lambda t)$ and T is independent of the process, then

$$E[N] = \lambda E[T],$$
$$\operatorname{Var}(N) = \lambda E[T] + \lambda^2 \operatorname{Var}(T),$$
$$\operatorname{Cov}(N, T) = \lambda \operatorname{Var}(T).$$

These follow by conditioning; they avoid finding the marginal law of N.

Overlapping intervals For counts from the same process on intervals I and J, split the timeline into disjoint pieces. Only the shared piece contributes to both counts, hence

$$\operatorname{Cov}(N(I), N(J)) = \lambda\, |I \cap J|.$$

This is the count analogue of overlapping indicators.

Conditioning can remove the rate Given $N(T) = n$, the original rate no longer matters for arrival positions. Thus a half-interval subcount is $\mathrm{Bin}(n, 1/2)$, and if exactly one arrival occurred, its time is uniform on the interval. If A is "at least one type-1 arrival" and each arrival is type 1 with probability p, then

$$P(A \mid N = n) = 1 - (1 - p)^n.$$

Use this likelihood with Bayes to find the conditional law of N given A.

6. Two-dimensional continuous variables: support first

The support picture controls every limit. A marginal is a *projection*: fix the required variable and integrate over the full allowed slice of the other. A conditional density is the normalized slice at the conditioning value, so its support usually changes with that value.

Reliable order: draw/describe S; find boundary intersections; normalize if needed; choose vertical or horizontal slices; write limits from the picture; integrate over $S \cap$ event; state zero outside the resulting support.

Marginal and conditional densities *Lookup:* צפיפות שולית *Lookup:* צפיפות מותנה

$$f_X(x) = \int_{\{y:(x,y)\in S\}} f(x, y)\, dy,$$
$$f_{Y\mid X=x}(y) = \frac{f(x, y)}{f_X(x)}.$$

The denominator is a density value, not $P(X = x)$. Verify the conditional density integrates to 1 on the fixed-x slice.

Vertical versus horizontal slices For $S = \{0 < y < x < 1\}$,

$$f_X(x) = \int_0^x f(x, y)dy, \qquad f_Y(y) = \int_y^1 f(x, y)dx.$$

The same region gives different limits because the outer variable is fixed in a different direction.

Uniform point in a region (2025). If $f = c$ on $0 < x < 1$, $1 - x < y < x + 2$, then area $= 2$, so $c = 1/2$. The vertical slice length is $(x + 2) - (1 - x) = 2x + 1$; therefore

$$f_X(x) = \tfrac{1}{2}(2x + 1) = x + \tfrac{1}{2}, \qquad 0 < x < 1.$$

For a uniform region, probabilities are area ratios

and marginals are density × slice length.

Curved boundaries and piecewise limits When reversing order, solve the boundary for the inner variable and intersect with all original bounds. For $y < x^2$: if $y < 0$, every real x satisfies the inequality;

if $y \ge 0$, then $x < -\sqrt{y}$ or $x > \sqrt{y}$. With an original restriction $x \ge 0$, only $x > \sqrt{y}$ remains. Breakpoints appear where candidate bounds cross or where a curve reaches an endpoint.

Q3 continued – curved-support template. If $1 < x < 2$ and $0 < y < x^2$, then $0 < y < 4$. For fixed y, x must satisfy $x > \sqrt{y}$ and $1 < x < 2$: use $1 < x < 2$ for $0 < y < 1$, but $\sqrt{y} < x < 2$ for $1 < y < 4$. The breakpoint $y = 1$ comes from the endpoint $x = 1$.

Independence and event regions If the density is positive on a support that is not a Cartesian product $S_X \times S_Y$, independence is impossible. If the support is a product, factorization $f = f_X f_Y$ must still be checked. Event probabilities require integrating over the intersection of the event with S; for conditional events, numerator uses both conditions and denominator uses the given condition. **Trap:** Factoring the algebra is not enough when the support couples x and y. Zero covariance does not generally imply independence.

Choosing integration order Choose the order that makes $S \cap A$ easiest to describe. If one order needs several pieces and the other needs one, switch. A safe recipe is to convert every condition into lower/upper bounds on the inner variable, then use $\max\{\text{lowers}\} < \min\{\text{uppers}\}$ and split where the winner changes.

7. Conditional expectation and covariance

Conditioning is especially valuable when $Y \mid X = x$ belongs to a familiar family. Then its conditional mean or second moment can be inserted before averaging:

$$E[Y] = E(E[Y \mid X]), \qquad E[Y^2] = E(E[Y^2 \mid X]),$$

$$\operatorname{Var}(Y) = E[Y^2] - E[Y]^2.$$

Ex: If $Y \mid X = x \sim \text{Poi}(10x)$, then $E[Y \mid X] = 10X$ and $E[Y] = 10E[X]$ without deriving f_Y. For joint moments,

$$E[g(X,Y)] = \iint_S g(x,y) f(x,y)\,dx\,dy,$$

and $\operatorname{Cov}(X,Y) = E[XY] - E[X]E[Y]$. *Lookup:* שונות משותפת Conditional symmetry can save work: if $X \mid Y = y$ is symmetric around 0, then $E[X \mid Y] = 0$ and odd conditional moments vanish.

When the conditional variance is easy, the decomposition

$$\operatorname{Var}(Y) = E[\operatorname{Var}(Y \mid X)] + \operatorname{Var}(E[Y \mid X])$$

separates randomness inside each case from randomness between cases. For $Y \mid X \sim \text{Poi}(\lambda X)$, both the conditional mean and variance are λX.

8. Sums, extrema and transformations

Which method? Independent sum with known densities → convolution; maximum/minimum → CDF events; one-variable transformation $Y = g(X)$ → CDF method when unsure; conditional model simple → condition rather than derive a difficult marginal.

Convolution *Lookup:* צפיפות סכום/הפרש For $Z = X + Y$, use the supplied formula, but first find the possible z-range. For fixed z, require both $x \in S_X$ and $z - x \in S_Y$; equivalently, integrate over $\max\{\text{lower bounds}\} < x < \min\{\text{upper bounds}\}$. Split where active endpoints change. **Ex:** If $X, Y \sim U(0,1)$ independently, then $\max(0, z-1) < x < \min(1, z)$, giving one expression on $0 < z < 1$ and another on $1 < z < 2$.

Extrema and transforms For iid X_i, $P(\max X_i \le t) = F(t)^n$ and $P(\min X_i > t) = [1 - F(t)]^n$. Differentiate only after the complete piecewise CDF. For $Y = g(X)$, write $F_Y(y) = P(g(X) \le y)$; if g is not one-to-one, split into inverse branches and add their contributions. Always derive the transformed support.

Known closure before calculation Independent Poisson variables add to a Poisson with summed parameters. Independent normals remain normal with means added linearly and variances multiplied by squared coefficients. Use these closures before convolution or MGF algebra. For a difference, variances still add under independence.

9. Approximation selector and normal workflow

Original binomial: current-course rare-event rule $n \ge 50$ and $np < 5 \to \text{Poi}(np)$; if $np \ge 5$ and $n(1-p) \ge 5$, use $N(np, np(1-p))$.
Original Poisson: for sufficiently large λ, use $N(\lambda, \lambda)$; the current material gives no numerical cutoff.
General iid sum/mean: check iid and finite mean/variance, compute one unit's moments, then CLT.

For every normal approximation: (1) name the exact original law; (2) compute $\mu = E[X]$, $\sigma^2 = \operatorname{Var}(X)$, σ; (3) define $Y \sim N(\mu, \sigma^2)$; (4) continuity-correct if the original variable is discrete; (5) standardize; (6) use the supplied Z-table.

$$P(X \le k) \approx P(Y \le k + 0.5),$$
$$P(X \ge k) \approx P(Y \ge k - 0.5).$$

The table gives $\Phi(z) = P(Z \le z)$; right tail $= 1 - \Phi(z)$, interval $= \Phi(b) - \Phi(a)$, and $\Phi(-z) = 1 - \Phi(z)$.

Poisson → normal. $X \sim \text{Poi}(60)$ and $P(X > 80) = P(X \ge 81)$. Let $Y \sim N(60, 60)$:

$$P(X > 80) \approx P(Y > 80.5) = P\left(Z > \frac{80.5 - 60}{\sqrt{60}}\right).$$

Binomial choices. If $X \sim \text{Bin}(1000, 0.002)$, the course rare-event rule supports $\text{Poi}(2)$.
If $X \sim \text{Bin}(60, 1/6)$, then $np = 10$ and $n(1-p) = 50$, so normal approximation is allowed: $Y \sim N(10, 50/6)$ and continuity correction is applied before standardizing.

Reading the supplied Z-table The table is a left-tail table. For $z = 2.65$, use row 2.6 and column .05 to get $\Phi(2.65)$. A right tail is $1 - \Phi(z)$; an interval is $\Phi(b) - \Phi(a)$. Standardize first and keep the inequality direction visible.

10. LLN, CLT and exact normal closure

The LLN tells where a long-run average settles: for iid variables satisfying the course's moment assumptions, $\bar{X}_n \to \mu$ in probability (and the strong law gives almost-sure convergence). It does not by itself give a numerical tail probability.
The CLT describes fluctuations around that value:

$$S_n \approx N(n\mu, n\sigma^2), \qquad \bar{X}_n \approx N(\mu, \sigma^2/n).$$

Define one repeated iid unit first. If $W_i = X_i - Y_i$, include the covariance inside one unit:

$$\operatorname{Var}(W_i) = \operatorname{Var} X_i + \operatorname{Var} Y_i - 2\operatorname{Cov}(X_i, Y_i).$$

Independent normal sums are *exactly* normal, so no large-n argument is needed; means combine linearly and variances use squared coefficients.

Repeated-pair CLT (2026). If (X_i, Y_i) are iid pairs but X_i and Y_i are dependent within a pair, define $W_i = X_i - Y_i$. The W_i are iid across i; compute $E[W_i]$ and $\operatorname{Var}(W_i)$ including $\operatorname{Cov}(X_i, Y_i)$, then apply the CLT to $\bar{W} = \bar{X} - \bar{Y}$. Do not approximate $\bar{X}$ and $\bar{Y}$ separately as if independent.

11. Moment-generating functions

An MGF packages all moments and often identifies a distribution. *Lookup:* פונקציה יוצרת מומנטים

$$M_X'(0) = E[X], \qquad M_X''(0) = E[X^2].$$

For independent sums only, $M_{X+Y} = M_X M_Y$; match the product to a known MGF. For $aX + b$, $M_{aX+b}(t) = e^{bt} M_X(at)$. MGFs require existence in a neighborhood of 0. **Trap:** Differentiate the MGF, not the log-MGF, to extract raw moments. **Ex:** If $M_S(t)$ factors into $[M_X(t)]^n$, ask whether S is a sum of n independent copies. If the product matches a Poisson, normal or gamma-family MGF from the supplied sheet, identify the distribution rather than differentiating a long expression.

12. Statistics: parameter, statistic, estimator, estimate

A **parameter** is an unknown fixed feature of the population. A **statistic** is a function of the random sample containing no unknown parameter. An **estimator** is a statistic chosen to estimate a parameter; after data are observed, its numerical value is the **estimate**. *Lookup:* סטטיסטיקה
For iid data, $E[\bar{X}] = \mu$ and $\operatorname{Var}(\bar{X}) = \sigma^2/n$. The unbiased sample variance is

$$S^2 = \frac{1}{n-1} \sum_i (X_i - \bar{X})^2.$$

Bias and MSE measure estimator quality:

$$\operatorname{Bias}(T) = E[T] - \theta,$$
$$\operatorname{MSE}(T) = \operatorname{Var}(T) + \operatorname{Bias}(T)^2.$$

Lookup: אומד חסר הטייה *Lookup:* תוחלת ריבוע הטעות של אומד

Comparing estimators Unbiasedness is not the only criterion. If T_1 is unbiased but variable and T_2 is slightly biased but concentrated, compare MSE. Always evaluate expectations under the full model and for every allowed parameter value.

13. Maximum likelihood estimation

MLE treats the observed data as fixed and varies the parameter: choose the parameter value under which those data are most plausible. *Lookup:* פונקציית נראות The log-likelihood is used because products become sums and log preserves the maximizer.

Regular versus support-driven MLE. Write the parameter domain and the full PMF/PDF including support indicators. First determine which parameter values make every observation possible. If support is fixed, maximize $\ell = \log L$ by differentiation and compare boundaries. If support depends on the parameter, the admissible boundary may determine the answer before differentiation.

Fast patterns:

$$X_i \sim \text{Ber}(p) : \hat{p} = \bar{X}, \quad X \sim \text{Bin}(m,p) : \hat{p} = X/m,$$

$$X_i \sim \text{Bin}(m,p) : \hat{p} = \bar{X}/m, \quad X_i \sim \text{Poi}(\lambda) : \hat{\lambda} = \bar{X},$$

$$X_i \sim \text{Exp}(\lambda)\ (\text{rate}) : \quad \hat{\lambda} = 1/\bar{X}.$$

Boundary/support MLE. If $X_i \sim U(0, \theta)$, the data require $\theta \ge x_{(n)} = \max x_i$. On that admissible set $L \propto \theta^{-n}$ decreases, so $\hat{\theta} = x_{(n)}$. The 2026 support example similarly gives a maximum absolute observation.

For iid $N(\mu, \sigma^2)$, if both parameters are unknown,

$$\hat{\mu} = \bar{X}, \qquad \hat{\sigma}^2_{\text{MLE}} = \frac{1}{n} \sum_i (X_i - \bar{X})^2.$$

The MLE denominator is n; the unbiased sample variance denominator is $n - 1$. MLE invariance:

$$\widehat{g(\theta)}_{\text{MLE}} = g(\hat{\theta}_{\text{MLE}}).$$

Trap: Never discard a support indicator involving the parameter; a critical point outside the allowed domain is not an estimator.

Constrained interior solution If ℓ is concave or unimodal and its unconstrained global maximizer is a, then over $\theta \in [L, U]$ the constrained maximizer is

$$\hat{\theta} = \min\{U, \max\{L, a\}\}.$$

Without concavity/unimodality, compare every feasible stationary point and both boundaries directly. Recent exams use this when a binomial success probability is an affine function of a constrained parameter.

MLE followed by bias/MSE A support-based MLE is often an extreme order statistic. For a continuous estimator, find its CDF and differentiate to obtain the density; for a discrete estimator, obtain its PMF from CDF differences or direct probabilities. Then compute $E[\hat{\theta}]$ and $E[\hat{\theta}^2]$, followed by $\text{MSE} = \text{Var} + \text{Bias}^2$. This combines extrema, transformations and estimation in one question.

14. Recurring exam moves

Random size/composition: condition first; a series may appear only in the final algebra. Complicated count: indicators for expectation, overlap classes for variance. Poisson: split a timeline into disjoint intervals/categories before superposition, thinning or conditioning. Two-dimensional density: support before algebra; projection for marginals, normalized slice for conditionals and intersection for events. MLE: domain and support before derivatives; compare boundaries globally.

Small but recurring algebra $\operatorname{Var}(aX + bY) = a^2 \operatorname{Var} X + b^2 \operatorname{Var} Y + 2ab \operatorname{Cov}(X,Y)$. Independence implies zero covariance, but not conversely. A density may exceed 1; a probability may not. Adding a constant changes the mean but not the variance.

15. Official-sheet Hebrew lookup

צפיפות שולית	marginal density
צפיפות מותנה	conditional density
פונקציית ההתפלגות המצטברת	CDF
שונות משותפת	covariance
פונקציה יוצרת מומנטים	MGF
משפט הגבול המרכזי	CLT
פונקציית נראות	likelihood
אומד חסר הטיה	unbiased estimator

16. Final sanity check

Support stated? PMF/PDF and conditional density normalize? Event intersected with the support? Units consistent? Dependence checked before adding variances? Continuity correction applied only to a discrete-to-normal approximation? Likelihood maximized only over allowed parameters? Probability in $[0,1]$, variance nonnegative, and piecewise answer completed with zero/one outside its support?